

# **Bankruptcy Prediction Using Machine Learning**

## **Abstract**

Our project is geared towards employing machine learning methodologies to predict bankruptcy. By leveraging data analysis and predictive modeling, we aim to forecast the probability of a company encountering financial insolvency.

## **Dataset Used**

The data were collected from the Taiwan Economic Journal for the years 1999 to 2009. Company bankruptcy was defined based on the business regulations of the Taiwan Stock Exchange.

The columns contain 95 financial ratios of the company out of which best features are to be used for the optimum prediction.

Link: <https://www.kaggle.com/datasets/fedesoriano/company-bankruptcy-prediction>

## **Methodology**

### **➤ Importing Libraries and Data:**

- Essential libraries such as pandas, numpy, and scikit-learn are imported for data manipulation, analysis, and machine learning.
- The dataset is loaded from a CSV file into a DataFrame (df), containing features and the target variable indicating bankruptcy status.

### **➤ Checking Data Quality:**

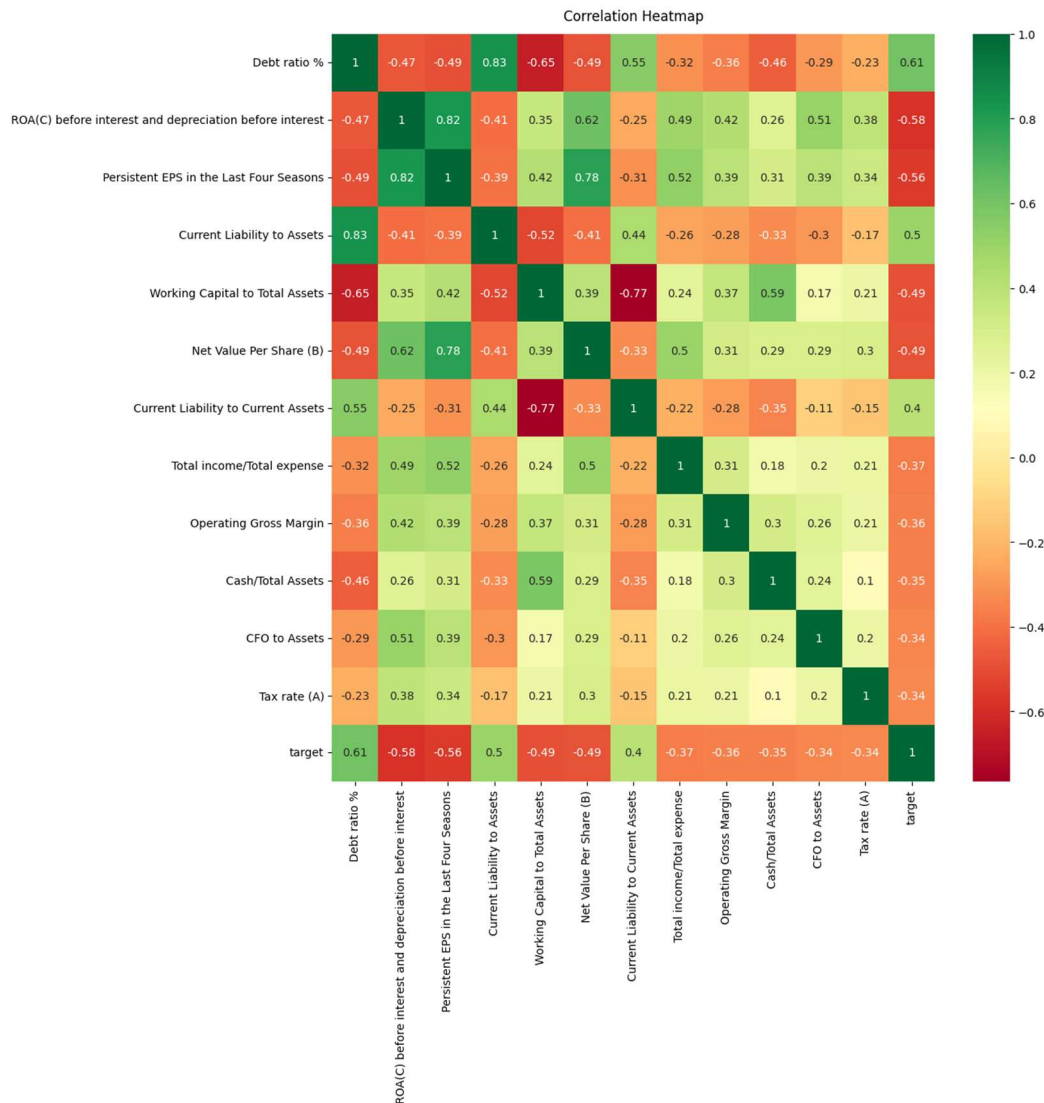
- Data quality is assessed by checking for missing values and duplicated rows using pandas' isna() and duplicated() functions, ensuring reliability for model training.

### **➤ Feature Engineering and Selection:**

- The target column is renamed to 'target' for clarity.
- Relevant features for predicting bankruptcy are identified through feature selection, enhancing model efficiency and interpretability.
- The SelectKBest method with the ANOVA F-value metric selects the top 25 best features based on their relationship with the target variable.

### **➤ Feature Reduction via Correlation Analysis:**

- A correlation heatmap is plotted for the top 25 selected features to identify those with a correlation above 0.85 as highly correlated.
- Redundant features are removed based on lower 'feature scores', retaining essential predictors.
- This leads to a final set of 13 features for training the bankruptcy prediction model.



➤ **Handling Class Imbalance:**

- Class imbalance is addressed using oversampling (SMOTE) technique to balance the dataset, preventing biased training.

➤ **Model Building and Evaluation:**

- Classification models including Decision Tree, Random Forest, Logistic Regression, Naive Bayes, SVM, and XGBoost are implemented.
- Models are evaluated on balanced and unbalanced datasets, with hyperparameter tuning performed via Bayesian optimization (BayesSearchCV).
- Performance metrics (accuracy, precision, recall, F1 score, AUC-ROC) and confusion matrices assess predictive capabilities.

➤ **Ensemble Modeling:**

- A Voting Classifier ensemble model combines multiple base models using soft voting, leveraging their collective strengths for improved accuracy.

➤ **Output and Interpretation:**

- Model performances are summarized in a DataFrame and saved as 'output.csv', with ensemble model metrics providing a comparative performance overview.

## **FINANCIAL RATIOS USED**

- **Net income to total assets** -  $\text{Net Income} = \text{Pre-Tax Income (EBT)} - \text{Taxes}$ .
- **Net worth/assets** - Net worth is calculated by subtracting all liabilities from assets.
- **Retained Earnings/Total Assets (RE/TA)** - This ratio measures the amount of reinvested earnings or losses, which reflects the extent of the company's leverage.
- **EPS** - Earnings per share (EPS) is calculated as a company's profit divided by the outstanding shares of its common stock. The resulting number serves as an indicator of a company's profitability.
- **Current Liabilities to Assets Ratio** - measures the proportion of a company's assets that are financed by its current liabilities, indicating short-term financial stability and the company's ability to meet its short-term obligations with its current assets.
- **The Working Capital to Total Assets ratio** - measures a company's ability to cover its short term financial obligations (Total Current Liabilities) by comparing its Total Current Assets to its Total Assets.
- **Current Ratio** - A firm's total current assets are divided by its total current liabilities. It shows the ability of a firm to meet its current liabilities with current assets.
- **Net income to stockholder's Equity** - Return on equity (ROE) is the measure of a company's net income divided by its shareholders' equity.
- **The Net Asset Value Per Share (NAVPS)** - is a real estate metric that indicates the value of a mutual fund or an exchange-traded fund (ETF). The NAVPS is obtained by dividing the net asset value (total assets less liabilities) of a fund by the number of outstanding shares.
- **Liabilities - assets flags** - the term "liabilities - assets" refers to the calculation where you subtract the total assets of a company from its total liabilities. This calculation can result in what is commonly referred to as "net liabilities."
- **Total expense/asset** - total expense ratio is calculated by dividing total fund costs by total fund assets. The total fund's costs are all costs associated with managing the operations of the investment. The total fund's assets are the total investments of the company in stocks, bonds, and securities.
- **Borrowing dependency** - Borrowing dependency in companies refers to the extent to which a company relies on borrowed funds, such as loans or bonds, to finance its operations or growth, indicating its leverage and potential financial risk exposure.
- **Working cap/equity** - The Working Capital to Equity ratio measures the proportion of shareholders' equity that is tied up in the day-to-day funds necessary to keep a company operating, providing insight into the company's operational efficiency and financial leverage.

## PERFORMANCE METRICS

- **Accuracy:** Accuracy is the percentage of predictions that the model makes correctly. It is calculated by dividing the number of correct predictions by the total number of predictions.
- **Precision:** Precision is the percentage of positive predictions that the model makes that are actually correct. It is calculated by dividing the number of true positives by the total number of positive predictions.
- **Recall:** Recall is the percentage of all positive examples that the model correctly identifies. It is calculated by dividing the number of true positives by the total number of positive examples.
- **F1 score:** The F1 score is a weighted average of precision and recall. It is calculated by taking the harmonic mean of precision and recall.
- **Confusion matrix:** A confusion matrix is a table that shows the number of predictions for each class, along with the actual class labels. It can be used to visualize the performance of the model and identify areas where the model is struggling.

## RESULTS

### ENSEMBLE MODEL RESULTS

In our project, we employed an ensemble approach to classification by combining multiple machine-learning algorithms using a Voting Classifier. This technique aggregates the predictions of individual classifiers and yields improved performance compared to any single classifier alone.

#### Ensemble Configuration

We assembled the following classification algorithms into our Voting Classifier:

- Random Forest Classifier
- Support Vector Machines (SVM)
- XG Boost

- Decision Trees
- Naive Bayes
- Logistic Regression

## **Results Summary**

Upon training and evaluating the ensemble classifier on our dataset, we obtained the following performance metrics:

```
Ensemble Model Metrics Unbalanced:  
Accuracy: 0.9340909090909091  
Precision: 0.901840490797546  
Recall: 0.9730142566191446  
F1 Score: 0.9360764144011756  
AUC-ROC: 0.9344029198927387  
Confusion Matrix:  
[[1788  208]  
 [  53 1911]]
```

## **Conclusion**

The ensemble classifier demonstrated robust performance across multiple evaluation metrics, indicating its effectiveness in accurately classifying instances in our dataset. By leveraging the diverse strengths of individual classifiers, the ensemble approach achieved superior predictive accuracy and generalization capability.